Comparison of relational database management systems

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The following tables compare general and technical information for a number of relational database management systems. Please see the individual products' articles for further information. Unless otherwise specified in footnotes, comparisons are based on the stable versions without any add-ons, extensions or external programs.

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General information

	Maintainer	First public release date	Latest stable version	Latest release date	License
4D (4th Dimension)	4D S.A.S.	1984	v15.2	2015-11-19[1]	Proprietary
ADABAS	Software AG	1970	8.1	2013-06	Proprietary
Adaptive Server Enterprise	SAP AG	1987	16.0	2015	Proprietary
Advantage Database Server (ADS)	SAP AG	1992	11.1	2012	Proprietary
Altibase	Altibase Corp.	2000	6.1.3	2014-04-18	Proprietary
Apache Derby	Apache	2004	10.13.1.1[2]	2016-10-29	Apache License
ClustrixDB	Clustrix	2010	v7.0	2015-08-19	Proprietary
CUBRID	NHN Corporation	2008-11	10.0.0	2016-02-05	GPL v2 or later
Datacom	CA, Inc.	Early 70s[3]	14[4]	2012[5]	Proprietary
DB2	IBM	1983	11.1	2016-04-12	Proprietary
Drizzle	Brian Aker	2008	7.2.4	2012-09-23	GPL v2 and v3, with some BSD components
Empress Embedded Database	Empress Software Inc	1979	10.20	2010-03	Proprietary
EXASolution	EXASOL AG	2004	4.2.8	2014-04-22	Proprietary
FileMaker	FileMaker, Inc., an Apple subsidiary	1985-04	15.0v2	2016-09-28	Proprietary
Firebird	Firebird project	2000-07-25	3.0.1	2016-09-27	IPL[6] and IDPL[7]
GPUdb	GIS Federal	2014	3.2.5	2015-01-14	Proprietary
HSQLDB	HSQL Development Group	2001	2.3.4	2016-05-16	BSD
H2	H2 Software	2005	1.3.176	2014-04-05	EPL and modified MPL
Informix Dynamic Server	IBM	1980	12.10.xC7	2016-06-15	Proprietary
Ingres	Ingres Corp.	1974	10.2	2014-09-30	GPL and Proprietary
InterBase	Embarcadero	1984	XE7 v12.0.4.357	2015-08-12	Proprietary
Linter SQL RDBMS	RELEX Group	1990	6.0.18, 6.1.15	2016-03-15	Proprietary
LucidDB	The Eigenbase Project	2007-01	0.9.3		GPL v2
MariaDB	MariaDB Community	2010-02-01	10.1.19	2016-11-07[8]	GPL v2 and LGPL for client-libraries
MaxDB	SAP AG	2003-05	7.9.0.8	2014	Proprietary
Microsoft Access (JET)	Microsoft	1992	16 (2016)	2015-09-22	Proprietary
Microsoft Visual Foxpro	Microsoft	1984	9 (2005)	2007-10-11	Proprietary
Microsoft SQL Server	Microsoft	1989	2016 (13)	2016-06-01	Proprietary
Microsoft SQL Server Compact (Embedded Database)	Microsoft	2000	2011 (v4.0)		Proprietary
MonetDB	The MonetDB Team / CWI	2004	Jun2016-SP2	2016-10-13	Mozilla Public License, version 2.0[9]
mSQL	Hughes Technologies	1994	3.11[10]	2012-Jun-01	Proprietary
MySQL	Oracle Corporation	1995-11	5.7.16[11]	2016-10-12	GPL v2 or Proprietary
MemSQL	MemSQL	2012-06	5.0.7 (2016)	2016-04-25	Proprietary
NexusDB	NexusDB Pty Ltd	2003	4.00.14	2015-06-25	Proprietary
HP NonStop SQL	Hewlett-Packard	1987	SQL/MX 2.3		Proprietary
Omnis Studio	TigerLogic Inc	1982-07	6.1.3 Release 1no	2015-12	Proprietary
OpenBase SQL	OpenBase International	1991	11.0.0		Proprietary
OpenEdge	Progress Software Corporation	1984	11.6.3	2016-08-19	Proprietary
OpenLink Virtuoso	OpenLink Software	1998	7.2.4	2016-04-24	GPL v2 or Proprietary
Oracle DB	Oracle Corporation	1979-11	12.1.0.2	2014-07-22[12]	Proprietary
Oracle Rdb	Oracle Corporation	1984	7.3.1.2	2014-10-08[13]	Proprietary
Paradox	Corel Corporation	1985	11	2003	Proprietary
Pervasive PSQL	Pervasive Software	1982	v12	2015	Proprietary
Polyhedra DBMS	ENEA AB	1993	9.0	2015-06-24	Proprietary, with Polyhedra Lite available as Freeware[14]
PostgreSQL	PostgreSQL Global Development Group	1989-06	9.6.1	2016-10-27[15]	PostgreSQL Licence (a liberal Open Source license)[16]
R:Base	R:BASE Technologies	1982	10.0	2016-05-26	Proprietary
Raima Database Manager	Raima Inc.	1984	14.0	2016-08-15	Proprietary
RDM Server	Raima Inc.	1993	8.4	2012-10-31	Proprietary
SAP HANA	SAP AG	2010	1.0 SPS12	2016-05-11	Proprietary
ScimoreDB	Scimore	2005	4.0.2782.26	2013-06-18	Proprietary

	Maintainer	First public release date	Latest stable version	Latest release date	License
SmallSQL	SmallSQL	2005-04-16	0.21	2011-06	LGPL
solidDB	UNICOM Global	1992	7.0.0.10	2014-04-29	Proprietary
SQL Anywhere	Sybase	1992	17.0	2015-07-15	Proprietary
SQLBase	Unify Corp.	1982	11.5	2008-11	Proprietary
SQLite	D. Richard Hipp	2000-09-12	3.15.1	2016-11-04[17]	Public domain
SQream DB	SQream Technologies	2014	1.9.9[18]	2016-10	Proprietary
Superbase	Superbase	1984	Scientific (2004)		Proprietary
Teradata	Teradata	1984	15	2014-04	Proprietary
Tibero	TmaxSoft	1992	6	2015-04	Proprietary
UniData	Rocket Software	1988	7.2.12	2011-10	Proprietary

Operating system support

The operating systems that the RDBMSes can run on.

	Windows	OS X	Linux	BSD	UNIX	AmigaOS	Symbian	z/OS	iOS	Android	OpenVMS
4th Dimension	Yes	Yes	No	No	No	No	No	No	No	No	No
ADABAS	Yes	No	Yes	No	Yes	No	No	Yes	No	No	No
Adaptive Server Enterprise	Yes	No	Yes	Yes	Yes	No	No	No	Yes	Yes	No
Advantage Database Server	Yes	No	Yes	No	No	No	No	No	No	No	No
Altibase	Yes	No	Yes	No	Yes	No	No	No	No	No	No
Apache Derby	Yes	Yes	Yes	Yes	Yes	No	No	Yes	?	No	No
ClustrixDB	No	No	Yes	No	Yes	No	No	No	No	No	No
CUBRID	Yes	Partial	Yes	No	No	No	No	No	No	No	No
Drizzle	No	Yes	Yes	Yes	Yes	No	No	No	No	No	No
DB2	Yes	Yes (Express C)	Yes	No	Yes	No	No	Yes	Yes	No	No
Empress Embedded Database	Yes	Yes	Yes	Yes	Yes	No	No	No	No	Yes	No
EXASolution	No	No	Yes	No	No	No	No	No	No	No	No
FileMaker	Yes	Yes	No	No	No	No	No	No	Yes	No	No
Firebird	Yes	Yes	Yes	Yes	Yes	No	No	Maybe	No	No	No
HSQLDB	Yes	Yes	Yes	Yes	Yes	No	No	Yes	?	?	No
H2	Yes	Yes	Yes	Yes	Yes	No	No	Yes	?	Yes	No
Informix Dynamic Server	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No	No
Ingres	Yes	Yes	Yes	Yes	Yes	No	No	Partial	No	No	?
InterBase	Yes	Yes	Yes	No	Yes (Solaris)	No	No	No	Yes	Yes	No
Linter SQL RDBMS	Yes	Yes	Yes	Yes	Yes	No	No	Under Linux on z Systems	Yes	Yes	Yes
LucidDB	Yes	Yes	Yes	No	No	No	No	No	No	No	No
MariaDB	Yes	Yes[19]	Yes	Yes	Yes	No	No	No	?	?	No
MaxDB	Yes	No	Yes	No	Yes	No	No	Maybe	No	No	No
Microsoft Access (JET)	Yes	No	No	No	No	No	No	No	No	No	No
Microsoft Visual Foxpro	Yes	No	No	No	No	No	No	No	No	No	No
Microsoft SQL Server	Yes	No	No	No	No	No	No	No	No	No	No
Microsoft SQL Server Compact (Embedded Database)	Yes	No	No	No	No	No	No	No	No	No	No
MonetDB	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No	No
MySQL	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	?	Yes[20]	No
Omnis Studio	Yes	Yes	Yes	No	No	No	No	No	No	No	No
OpenBase SQL	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No	No
OpenEdge	Yes	No	Yes	No	Yes	No	No	No	No	No	No
OpenLink Virtuoso	Yes	Yes	Yes	Yes	Yes	No	No	Yes	No	No	No
Oracle	Yes	Yes	Yes	No	Yes	No	No	Yes	No	No	Yes
Oracle Rdb	No	No	No	No	No	No	No	No	No	No	Yes
Pervasive PSQL	Yes	Yes (OEM only)	Yes	No	No	No	No	No	No	No	No
Polyhedra	Yes	No	Yes	No	Yes	No	No	No	No	No	No
PostgreSQL	Yes	Yes	Yes	Yes	Yes	No	No	Under Linux on z Systems[21]	No	Yes	No
R:Base	Yes	No	No	No	No	No	No	No	No	No	No
Raima Database Manager	Yes	Yes	Yes	Yes	Yes	No	No	No	Yes	Yes	No
RDM Server	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No	No
SAP HANA	No	No	Yes	No	No	No	No	No	No	No	No
ScimoreDB	Yes	No	No	No	No	No	No	No	No	No	No
SmallSQL	Yes	Yes	Yes	Yes	Yes	No	No	Yes	No	No	No
solidDB	Yes	No	Yes	No	Yes	No	No	Under Linux on z Systems	No	No	No
SQL Anywhere	Yes	Yes	Yes	No	Yes	No	No	No	No	Yes	No
SQLBase	Yes	No	Yes	No	No	No	No	No	No	No	No
SQLite	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Maybe	Yes	Yes	No
SQream DB	No	No	Yes	No	No	No	No	No	No	No	No
Superbase	Yes	No	No	No	No	Yes	No	No	No	No	No
Teradata	Yes	No	Yes	No	Yes	No	No	No	No	No	No
Tibero	Yes	No	Yes	No	Yes	No	No	No	No	No	No
UniData	Yes	No	Yes	No	Yes	No	No	No	No	No	No
UniVerse	Yes	No	Yes	No	Yes	No	No	No	No	No	No

Fundamental features

Information about what fundamental RDBMS features are implemented natively.

Database Name	ACID	Referential integrity	Transactions	Fine-grained locking	Multiversion concurrency control	Unicode	Interface
4th Dimension	Yes	Yes	Yes	?	?	Yes	GUI & SQL
ADABAS	Yes	No	Yes	?	?	Yes	proprietary direct call & SQL (via 3rd party)
Adaptive Server Enterprise	Yes	Yes	Yes	?	?	Yes	SQL
Advantage Database Server	Yes	Yes	Yes	Yes (Row-level locking)	?	Yes[4]	API & SQL
Altibase	Yes	Yes	Yes	Yes (Row-level locking)	?	Yes	API & GUI & SQL
Apache Derby	Yes	Yes	Yes	?	?	Yes	SQL
ClustrixDB	Yes	Yes	Yes	?	?	Yes	SQL
CUBRID	Yes	Yes	Yes	Yes (Row-level locking)	?	Yes	GUI & SQL
Drizzle	Yes	Yes	Yes	?	?	Yes	SQL
DB2	Yes	Yes	Yes	Yes (Row-level locking)[22]	?	Yes	GUI & SQL
Empress Embedded Database	Yes	Yes	Yes	?	?	Yes	API & SQL
EXASolution	Yes	Yes	Yes	?	?	Yes	API & GUI & SQL
Firebird	Yes	Yes	Yes	?	Yes	Yes	SQL
HSQLDB	Yes	Yes	Yes	?	Yes	Yes	SQL
H2	Yes	Yes	Yes	?	Yes[23]	Yes	SQL
Informix Dynamic Server	Yes	Yes	Yes	Yes (Row-level locking)	Yes	Yes	SQL and JSON
Ingres	Yes	Yes	Yes	?	?	Yes	SQL & QUEL
InterBase	Yes	Yes	Yes	?	?	Yes	SQL
Linter SQL RDBMS	Yes	Yes	Yes (Except for DDL)	Yes (Row-level locking)	?	Yes	API & GUI & SQL
LucidDB	Yes	No	No	?	?	Yes	SQL
MariaDB	Yes[2]	Yes[6]	Yes[2] except for DDL[24]	Yes (Row-level locking)	Yes	Yes	SQL
MaxDB	Yes	Yes	Yes	?	?	Yes	SQL
Microsoft Access (JET)	Yes	Yes	Yes	?	?	Yes	GUI & SQL
Microsoft Visual FoxPro	No	Yes	Yes	?	?	No	GUI & SQL
Microsoft SQL Server	Yes	Yes	Yes	Yes (Row-level locking)[25]	?	Yes	GUI & SQL
Microsoft SQL Server Compact (Embedded Database)	Yes	Yes	Yes	?	?	Yes	GUI & SQL
MonetDB	Yes	Yes	Yes	?	?	Yes	SQL
MySQL	Yes[2]	Yes[3]	Yes[2] except for DDL[24]	Yes (Row-level locking)[26]	Yes	Yes	GUI [5] & SQL
OpenBase SQL	Yes	Yes	Yes	?	?	Yes	GUI & SQL
Oracle	Yes	Yes	Yes except for DDL[24]	Yes (Row-level locking)[27]	Yes	Yes	API & GUI & SQL
Oracle Rdb	Yes	Yes	Yes	?	?	Yes	SQL
OpenLink Virtuoso	Yes	Yes	Yes	?	?	Yes	API & GUI & SQL
Pervasive PSQL	Yes	Yes	Yes	?	?	Yes[6]	API & GUI & SQL
Polyhedra DBMS	Yes	Yes	Yes	Yes (optimistic and pessimistic cell-level locking)[28]	?	Yes	API & SQL
PostgreSQL	Yes	Yes	Yes	Yes (Row-level locking)[29]	Yes	Yes	API & GUI & SQL
Raima Database Manager	Yes	Yes	Yes	?	?	Yes	SQL & API
RDM Server	Yes	Yes	Yes	?	?	Yes	SQL & API
SAP HANA	Yes	Yes	Yes	Yes (Row-level locking)	Yes	Yes	API & GUI & SQL
ScimoreDB	Yes	Yes	Yes	?	?	Partial	SQL
solidDB	Yes	Yes	Yes	Yes (Row-level locking)	?	Yes	API & SQL
SQL Anywhere	Yes	Yes	Yes	?	?	Yes	SQL

Database Name	ACID	Referential integrity	Transactions	Fine-grained locking	Multiversion concurrency control	Unicode	Interface
SQLBase	Yes	Yes	Yes	?	?	Yes	API & GUI & SQL
SQLite	Yes	Yes	Yes	No (Database-level locking)[30]	No	Optional[31]	API & SQL
Teradata	Yes	Yes	Yes	Yes (Hash and Partition)	?	Yes	SQL
Tibero	Yes	Yes	Yes	Yes (Row-level locking)	Yes	Yes	API & GUI & SQL
UniData	Yes	No	Yes	?	?	Yes	Multiple
UniVerse	Yes	No	Yes	?	?	Yes	Multiple
Database Name	**ACID**	**Referential integrity**	**Transactions**	**Fine-grained locking**	**Multiversion concurrency control**	**Unicode**	**Interface**

Note (1): Currently only supports read uncommited transaction isolation. Version 1.9 adds serializable isolation and version 2.0 will be fully ACID compliant.

Note (2): MySQL provides ACID compliance through the default InnoDB storage engine.[32][33]

Note (3): "For other than InnoDB storage engines, MySQL Server parses and ignores the FOREIGN KEY and REFERENCES syntax in CREATE TABLE statements. The CHECK clause is parsed but ignored by all storage engines."[34]

Note (4): Support for Unicode is new in version 10.0.

Note (5): MySQL provides GUI interface through MySQL Workbench.

Note (6): MariaDB's default XtraDB engine is ACID compliant.

Limits

Information about data size limits.

	Max DB size	Max table size	Max row size	Max columns per row	Max Blob/Clob size	Max CHAR size	Max NUMBER size	Min DATE value	Max DATE value	Max column name size
4th Dimension	Limited	?	?	65,135	200 GB (2 GiB Unicode)	200 GB (2 GiB Unicode)	64 bits	?	?	?
Advantage Database Server	Unlimited	16 EiB	65,530 B	65,135 / (10+ AvgFieldNameLength)	4 GiB	?	64 bits	?	?	128
Apache Derby	Unlimited	Unlimited	Unlimited	1,012 (5,000 in views)	2,147,483,647 chars	254 (VARCHAR: 32,672)	64 bits	0001-01-01	9999-12-31	128
ClustrixDB	Unlimited	Unlimited	64 MB on Appliance, 4 MB on AWS	?	64 MB	64 MB	64 MB	0001-01-01	9999-12-31	254
CUBRID	2 EB	2 EB	Unlimited	Unlimited	Unlimited	1 GB	64 bits	0001-01-01	9999-12-31	254
Drizzle	Unlimited	64 TB	8 KB	1,000	4 GB (longtext, longblob)	64 KB (text)	64 bits	0001	9999	64
DB2	Unlimited	2 ZB	32,677 B	1,012	2 GB	32 KiB)	64 bits	0001-01-01	9999-12-31	128
Empress Embedded Database	Unlimited	2^{63}-1 bytes	2 GB	32,767	2 GB	2 GB	64 bits	0000-01-01	9999-12-31	32
EXASolution	Unlimited	Unlimited	Unlimited	10,000	N/A	2 MB	128 bits	0001-01-01	9999-12-31	256
FileMaker	8 TB	8 TB	8 TB	256,000,000	4 GB	10,000,000	1 billion characters, 10^-400 to 10^400, +-	0001-01-01	4000-12-31	100
Firebird	Unlimited[1]	~32 TB	65,536 B	Depends on data types used	2 GB	32,767 B	64 bits	100	32768	31
HSQLDB	64 TB	Unlimited[8]	Unlimited[8]	Unlimited[8]	64 TB[7]	Unlimited[8]	Unlimited[8]	0001-01-01	9999-12-31	128
H2	64 TB	Unlimited[8]	Unlimited[8]	Unlimited[8]	64 TB[7]	Unlimited[8]	64 bits	-99999999	99999999	Unlimited[8]
	Max DB size	**Max table size**	**Max row size**	**Max columns per row**	**Max Blob/Clob size**	**Max CHAR size**	**Max NUMBER size**	**Min DATE value**	**Max DATE value**	**Max column name size**
Informix Dynamic Server	~128 PB	~128 PB	32,765 bytes (exclusive of large objects)	32,765	4 TB	32,765	10^{32}	01/01/0001[10]	12/31/9999	128 bytes
Ingres	Unlimited	Unlimited	256 KB	1,024	2 GB	32 000 B	64 bits	0001	9999	256
InterBase	Unlimited[1]	~32 TB	65,536 B	Depends on data types used	2 GB	32,767 B	64 bits	100	32768	31
Linter SQL RDBMS	Unlimited	2^{30} rows	64 KB (w/o BLOBs), 4 GB (BLOB)	250	4 GB	4 KB	64 bits	0001-01-01	9999-12-31	66
MariaDB	Unlimited	MyISAM storage limits: 256 TB; Innodb storage limits: 64 TB; Aria storage limits: ???	64 KB[3]	4,096[4]	4 GB (longtext, longblob)	64 KB (text)	64 bits	1000	9999	64[35]
Microsoft Access (JET)	2 GB	2 GB	16 MB	255	64 KB (memo field), 1 GB ("OLE Object" field)	255 B (text field)	32 bits	0100	9999	64
Microsoft Visual Foxpro	Unlimited	2 GB	65,500 B	255	2 GB	16 MB	32 bits	0001	9999	10
Microsoft SQL Server	524,272 TB (32 767 files * 16 TB max file size)	524,272 TB	8,060 bytes/2TB[6]	1,024/30,000(with sparse columns)	2 GB/Unlimited (using RBS/FILESTREAM object)	2 GB[6]	126 bits[2]	0001	9999	128
Microsoft SQL Server Compact (Embedded Database)	4 GB	4 GB	8,060 bytes	1024	2 GB	4000	154 bits	0001	9999	128
MonetDB	Unlimited	Unlimited	Unlimited	Unlimited	Unlimited	Unlimited	64 bits	0	9999-12-31	?
MySQL	Unlimited	MyISAM storage limits: 256 TB; Innodb storage limits: 64 TB	64 KB[3]	4,096[4]	4 GB (longtext, longblob)	64 KB (text)	64 bits	1000	9999	64
OpenLink Virtuoso	32 TB per instance (Unlimited via elastic cluster)	DB size (or 32 TB)	4 KB	200	2 GB	2 GB	2^{31}	0	9999	100
Oracle	Unlimited (4 GB * block size per tablespace)	4 GB * block size (with BIGFILE tablespace)	8 KB	1,000	128 TB	32,767 B[11]	126 bits	−4712	9999	30
	Max DB size	**Max table size**	**Max row size**	**Max columns per row**	**Max Blob/Clob size**	**Max CHAR size**	**Max NUMBER size**	**Min DATE value**	**Max DATE value**	**Max column name size**
Pervasive PSQL	4 billion objects	256 GB	2 GB	1,536	2 GB	8,000 bytes	64 bits	01-01-0001	12-31-9999	128 bytes
Polyhedra	Limited by available RAM, address space	2^{32} rows	Unlimited	65,536	4 GB (subject to RAM)	4 GB (subject to RAM)	64 bits	0001-01-01	8000-12-31	255
PostgreSQL	Unlimited	32 TB	1.6 TB	250–1600 depending on type	1 GB (text, bytea)[36] - stored inline or 4 TB (stored in pg_largeobject)[37]	1 GB	Unlimited	−4,713	5,874,897	63

	Max DB size	Max table size	Max row size	Max columns per row	Max Blob/Clob size	Max CHAR size	Max NUMBER size	Min DATE value	Max DATE value	Max column name size
Raima Database Manager	Unlimited	2^{48}-1 rows	32 KB	1,000	4 GB	char: 256, varchar: 4 KB	64 bits	0001-01-01	11758978-12-31	31
RDM Server	Unlimited	2^{64}-1 rows	32 KB	32,768	Unlimited	32 KB	64 bits	0001-01-01	11758978-12-31	32
SAP HANA	?	?	?	?	?	?	?	?	?	?
ScimoreDB	Unlimited	16 EB	8,050 B	255	16 TB	8,000 B	64 bits	?	?	?
solidDB	256 TB	256 TB	32 KB + BLOB data	Limited by row size	4 GB	4 GB	64 bits	-32768-01-01	32767-12-31	254
SQL Anywhere	104 TB (13 files, each file up to 8 TB (32 KB pages))	Limited by file size	Limited by file size	45,000	2 GB	2 GB	64 bits	0001-01-01	9999-12-31	?
SQLite	128 TB (2^{31} pages * 64 KB max page size)	Limited by file size	Limited by file size	32,767	2 GB	2 GB	64 bits	No DATE type[9]	No DATE type[9]	Unlimited
Teradata	Unlimited	Unlimited	64000 wo/lobs (64 GB w/lobs)	2,048	2 GB	64,000	38 digits	0001-01-01	9999-12-31	128
UniVerse	Unlimited	Unlimited	Unlimited	Unlimited	Unlimited	Unlimited	Unlimited	Unlimited	Unlimited	Unlimited
	Max DB size	**Max table size**	**Max row size**	**Max columns per row**	**Max Blob/Clob size**	**Max CHAR size**	**Max NUMBER size**	**Min DATE value**	**Max DATE value**	**Max column name size**

Note (1): Firebird 2.x maximum database size is effectively unlimited with the largest known database size >980 GB.[38] Firebird 1.5.x maximum database size: 32 TB.

Note (2): Limit is 10^{38} using `DECIMAL` datatype.[39]

Note (3): InnoDB is limited to 8,000 bytes (excluding `VARBINARY`, `VARCHAR`, `BLOB`, or `TEXT` columns).[40]

Note (4): InnoDB is limited to 1,000 columns.[40]

Note (6): Using `VARCHAR (MAX)` in SQL 2005 and later.[41]

Note (7): When using a page size of 32 KB, and when BLOB/CLOB data is stored in the database file.

Note (8): Java array size limit of 2,147,483,648 (2^{31}) objects per array applies. This limit applies to number of characters in names, rows per table, columns per table, and characters per `CHAR`/`VARCHAR`.

Note (9): Despite the lack of a date datatype, SQLite does include date and time functions,[42] which work for timestamps between 24 November 4714 B.C. and 1 November 5352.

Note (10): Informix DATETIME type has adjustable range from YEAR only through 1/10000th second. DATETIME date range is 0001-01-01 00:00:00.00000 through 9999-12-31 23:59:59.99999.

Note (11): Since version 12c. Earlier versions support up to 4000 B.

Tables and views

Information about what tables and views (other than basic ones) are supported natively.

	Temporary table	Materialized view
4th Dimension	Yes	No
ADABAS	?	?
Adaptive Server Enterprise	Yes[1]	Yes - see precomputed result sets
Advantage Database Server	Yes	No (only common views)
Altibase	Yes	No (only common views)
Apache Derby	Yes	No
ClustrixDB	Yes	No
CUBRID	No	No
Drizzle	Yes	No[4]
DB2	Yes	Yes
Empress Embedded Database	Yes	Yes
EXASolution	Yes	No
Firebird	Yes	No (only common views)
HSQLDB	Yes	No
H2	Yes	No (only common views)
Informix Dynamic Server	Yes	No[2]
Ingres	Yes	No
InterBase	Yes	No
Linter SQL RDBMS	Yes	Yes
LucidDB	No	No
MariaDB	Yes	No[4]
MaxDB	Yes	No
Microsoft Access (JET)	No	No
Microsoft Visual Foxpro	Yes	Yes
Microsoft SQL Server	Yes	Yes
Microsoft SQL Server Compact (Embedded Database)	Yes	No
MonetDB	Yes	No
MySQL	Yes	No[4]
OpenBase SQL	Yes	Yes
Oracle	Yes	Yes
Oracle Rdb	Yes	Yes
OpenLink Virtuoso	Yes	Yes
Pervasive PSQL	Yes	No
Polyhedra DBMS	No	No (only common views)
PostgreSQL	Yes	Yes[5]
Raima Database Manager	Yes	No
RDM Server	Yes	No
SAP HANA	?	?
solidDB	Yes	No (only common views)
SQL Anywhere	Yes	Yes
ScimoreDB	No	No
SQLite	Yes	No
Teradata	Yes	Yes
UniData	Yes	No
UniVerse	Yes	No

Note (1): Server provides tempdb, which can be used for public and private (for the session) temp tables.[43]

Note (2): Materialized views are not supported in Informix; the term is used in IBM's documentation to refer to a temporary table created to run the view's query when it is too complex, but one cannot for example define the way it is refreshed or build an index on it. The term is defined in the Informix Performance Guide.[44]

Note (4): Materialized views can be emulated using stored procedures and triggers.[45]

Note (5): Materialized views are now standard but can be emulated in versions prior to 9.3 with stored procedures and triggers using PL/pgSQL, PL/Perl, PL/Python, or other procedural languages.[46]

Indexes

Information about what indexes (other than basic B-/B+ tree indexes) are supported natively.

	R-/R+ tree	Hash	Expression	Partial	Reverse	Bitmap	GiST	GIN	Full-text	Spatial	FOT
4th Dimension	?	Cluster	?	?	?	?	?	?	Yes	?	?
ADABAS	?	?	?	?	?	?	?	?	?	?	?
Adaptive Server Enterprise	No	No	Yes	No	Yes	No	No	No	Yes	?	?
Advantage Database Server	No	No	Yes	No	Yes	Yes	No	No	Yes	?	?
Apache Derby	No	No	No	No	No	No	No	No	No[47]	?	?
ClustrixDB	No	Yes	No	No	No	No	No	No	No	No	?
CUBRID	No	No	Yes[48]	Yes[48]	Yes	No	No	No	No	No	No
Drizzle	No	No	No	No	No	No	No	No	No	?	?
DB2	No	?	Yes	No	Yes	Yes	No	No	Yes[49]	?	?
Empress Embedded Database	Yes	No	No	Yes	No	Yes	No	No	No	?	?
EXASolution	No	Yes	No	No	No	No	No	No	No	?	?
Firebird	No	No	Yes	No	Yes	No	No	No	No[50]	?	?
HSQLDB	No	No	No	No	No	No	No	No	No	?	?
H2	No	Yes	No	No	No	No	No	No	Yes[51]	Yes[52]	?
Informix Dynamic Server	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes[53]
Ingres	Yes	Yes	Ingres v10	No	No	Ingres v10	No	No	No	?	?
InterBase	No	No	No	No	No	No	No	No	No	?	?
Linter SQL RDBMS[10]	No	Yes temporary indexes for equality joins	Yes for some scalar functions like LOWER and UPPER	No	No	No	No	No	Yes[54]	Yes[55]	No
LucidDB	No	No	No	No	No	Yes	No	No	No	?	?
MariaDB	Aria and MyISAM tables only[56]	MEMORYy,[56] InnoDB,[5] tables only	PERSISTENT virtual columns only [57]	No	No	No	No	No	Yes[58]	MyISAM tables only[59]	?
MaxDB	No	No	No	No	No	No	No	No	No	?	?
Microsoft Access (JET)	No	No	No	No	No	No	No	No	No[60]	?	?
Microsoft Visual Foxpro	No	No	Yes	Yes	Yes[2]	Yes	No	No	No	?	?
Microsoft SQL Server	Spatial Indexs	Yes[4]	Yes[3]	Yes	on Computed columns[3]	Bitmap filter index for Star Join Query	No	No	Yes[61]	Yes[62]	?
Microsoft SQL Server Compact (Embedded Database)	No	No	No	No	No	No	No	No	No[63]	?	?
MonetDB	No	Yes	No	No	No	Yes	No	No	No	No	No
MySQL	MyISAM tables only	MEMORY, Cluster (NDB), InnoDB,[5] tables only	No[64]	No	No	No	No	No	MyISAM tables[65] and, since v5.6.4, InnoDB tables[66]	MyISAM tables only[59]	?
Oracle	Yes [11]	Cluster Tables	Yes	Yes [6]	Yes	Yes	No	No	Yes[67]	Yes[68]	?
Oracle Rdb	No	Yes	?	No	No	?	No	No	?	?	?
OpenLink Virtuoso	Yes	Cluster	Yes	Yes	No	Yes	No	No	Yes	Yes (Commercial only)	No
Pervasive PSQL	No	No	No	No	No	No	No	No	No	No	No
Polyhedra DBMS	No	Yes	No	No	No	No	No	No	No	No	?
PostgreSQL	Yes	Yes	Yes	Yes	Yes[7]	Yes	Yes	Yes	Yes[69]	PostGIS[70]	?
Raima Database Manager	No	Yes	No	Yes	Yes	No	No	No	No	No	No
RDM Server	No	No	No	Yes	Yes	No	No	No	No	No	No
SAP HANA	?	?	?	?	?	?	?	?	?	?	?
ScimoreDB	No	No	No	No	No	No	No	No	Yes[71]	?	?
solidDB	No	No	No	No	Yes	No	No	No	No	No	No
SQL Anywhere	No	No	No	No	No	No	No	No	Yes	Yes	?
SQLite	Yes[72]	No	Yes[73]	Yes	No	No	No	No	Yes[74]	SpatiaLite[75]	?
SQream DB	?	?	?	?	?	?	?	?	?	?	?
Teradata	No	Yes	Yes	Yes	No	Yes	No	No	?[76]	?	?
UniVerse	Yes	Yes	Yes[3]	Yes[3]	Yes[3]	No	No	No	?	Yes[77]	?
	R-/R+ tree	**Hash**	**Expression**	**Partial**	**Reverse**	**Bitmap**	**GiST**	**GIN**	**Full-text**	**Spatial**	**FOT**

Note (1): The users need to use a function from freeAdhocUDF library or similar.[78]

Note (2): Can be implemented for most data types using expression-based indexes.

Note (3): Can be emulated by indexing a computed column[79] (doesn't easily update) or by using an "Indexed View"[80] (proper name not just any view works[81]).

Note (4): Used for InMemory ColumnStore index, temporary hash index for hash join, Non/Cluster & fill factor.

Note (5): InnoDB automatically generates adaptive hash index[82] entries as needed.

Note (6): Can be implemented using Function-based Indexes in Oracle 8i and higher, but the function needs to be used in the sql for the index to be used.

Note (7): A PostgreSQL functional index can be used to reverse the order of a field.

Note (10): B+ tree and full-text only for now.

Note (11): R-Tree indexing available in base edition with Locator but some functionality requires Personal Edition or Enterprise Edition with Spatial option.

Database capabilities

	Union	Intersect	Except	Inner joins	Outer joins	Inner selects	Merge joins	Blobs and Clobs	Common Table Expressions	Windowing Functions	Parallel Query
4th Dimension	Yes	Yes	Yes	Yes	Yes	No	No	Yes	?	?	?
ADABAS	Yes	?	?	?	?	?	?	?	?	?	?
Adaptive Server Enterprise	Yes	?	?	Yes	Yes	Yes	Yes	Yes	?	?	Yes
Advantage Database Server	Yes	No	No	Yes	Yes	Yes	Yes	Yes	?	No	?
Altibase	Yes	Yes	Yes, via MINUS	Yes	Yes	Yes	Yes	Yes	No	No	No
Apache Derby	Yes	Yes	Yes	Yes	Yes	Yes	?	Yes	No	No	?
ClustrixDB	Yes	No	No	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
CUBRID	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes[48]	?
Drizzle	Yes	No	No	Yes	Yes	Yes	No	Yes	No	No	No[83]
DB2	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes[84]
Empress Embedded Database	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	?	?	?
EXASolution	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes
Firebird	Yes	?	?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	?
HSQLDB	Yes	Yes	Yes	Yes	Yes	Yes	Yes[85]	Yes	Yes	No	Yes[85]
H2	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	experimental[86]	No[87]	?
Informix Dynamic Server	Yes	Yes	Yes, via MINUS	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes[88]
Ingres	Yes	No	No	Yes	Yes	Yes	Yes	Yes	No	No	?
InterBase	Yes	?	?	Yes	Yes	?	?	Yes	?	?	?
Linter SQL RDBMS	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No
LucidDB	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	?	?	?
MariaDB	Yes	No	No	Yes	Yes	Yes	No	Yes	No[89]	Yes[90]	No[91]
MaxDB	Yes	?	?	Yes	Yes	Yes	No	Yes	?	?	?
Microsoft Access (JET)	Yes	No	No	Yes	Yes	Yes	No	Yes	No	No	?
Microsoft Visual Foxpro	Yes	?	?	Yes	Yes	Yes	?	Yes	?	?	?
Microsoft SQL Server	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes[92]	Yes[93]
Microsoft SQL Server Compact (Embedded Database)	Yes	No	No	Yes	Yes	?	No	Yes	No	No	?
MonetDB	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
MySQL	Yes	No	No	Yes	Yes	Yes	No	Yes	No[89]	No	No[91]
OpenBase SQL	No	No	No	Yes	Yes	Yes	Yes	Yes	?	?	?
OpenLink Virtuoso	Yes	?	?	Yes	Yes	Yes	?	Yes	?	?	?
Oracle	Yes	Yes	Yes, via MINUS	Yes	Yes	Yes	Yes	Yes	Yes [1]	Yes	Yes[94]
Oracle Rdb	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	?	?	?
Pervasive PSQL	Yes	No	No	Yes	Yes	?	?	Yes	No	No	No
Polyhedra DBMS	Yes	Yes	Yes	Yes	Yes	No	No	Yes	No	No	No
PostgreSQL	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes[95]
Raima Database Manager	No	No	No	Yes	Yes	No	No	Yes	No	No	No
RDM Server	Yes	No	No	Yes	Yes	Yes	No	Yes	No	No	No
SAP HANA	?	?	?	?	?	?	?	?	?	?	?
ScimoreDB	Yes	?	?	Yes	LEFT only	Yes	Yes	Yes	?	?	?
SmallSQL	?	?	?	?	?	?	?	?	?	?	?
solidDB	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No
SQL Anywhere	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
SQLite	Yes	Yes	Yes	Yes	LEFT only	Yes	No	Yes	3.8.3+[96]	No	No
SQream DB	ALL only	No	No	Yes	Yes	Yes	Yes	No	Yes	Yes	No
Teradata	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
UniVerse	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	?
	Union	**Intersect**	**Except**	**Inner joins**	**Outer joins**	**Inner selects**	**Merge joins**	**Blobs and Clobs**	**Common Table Expressions**	**Windowing Functions**	**Parallel Query**

Note (1): Recursive CTEs introduced in 11gR2 supersedes similar construct called CONNECT BY.

Data types

	Type system	Integer	Floating point	Decimal	String	Binary	Date/Time	Boolean	Other
4th Dimension	Static	UUID (16-bit), SMALLINT (16-bit), INT (32-bit), BIGINT (64-bit), NUMERIC (64-bit)	REAL, FLOAT	REAL, FLOAT	CLOB, TEXT, VARCHAR	BIT, BIT VARYING, BLOB	DURATION, INTERVAL, TIMESTAMP	BOOLEAN	PICTURE
Altibase[97]	Static	SMALLINT (16-bit), INTEGER (32-bit), BIGINT (64-bit)	REAL (32-bit), DOUBLE (64-bit)	DECIMAL, NUMERIC, NUMBER, FLOAT	CHAR, VARCHAR, NCHAR, NVARCHAR, CLOB	BLOB, BYTE, NIBBLE, BIT, VARBIT	DATE		GEOMETR
ClustrixDB[98]	Static	TINYINT (8-bit), SMALLINT (16-bit), MEDIUMINT (24-bit), INT (32-bit), BIGINT (64-bit)	FLOAT (32-bit), DOUBLE	DECIMAL	CHAR, BINARY, VARCHAR, VARBINARY, TEXT, TINYTEXT, MEDIUMTEXT, LONGTEXT	TINYBLOB, BLOB, MEDIUMBLOB, LONGBLOB	DATETIME, DATE, TIMESTAMP, YEAR	BIT(1), BOOLEAN	ENUM, SE
CUBRID[99]	Static	SMALLINT (16-bit), INTEGER (32-bit), BIGINT (64-bit)	FLOAT, REAL(32-bit), DOUBLE(64-bit)	DECIMAL, NUMERIC	CHAR, VARCHAR, NCHAR, NVARCHAR, CLOB	BLOB	DATE, DATETIME, TIME, TIMESTAMP	BIT	MONETARY, BIT V SET, MULTISET, SE ENUM
Drizzle[100]	Static	INT (32-bit), BIGINT (64-bit)	DOUBLE (aka REAL) (64-bit)	DECIMAL	BINARY, VARCHAR, VARBINARY, TEXT,	BLOB	DATETIME, DATE, TIMESTAMP		ENUM, SER
Empress Embedded Database	Static	TINYINT, SQL_TINYINT, or INTEGER8; SMALLINT, SQL_SMALLINT, or INTEGER16; INTEGER, INT, SQL_INTEGER, or INTEGER32; BIGINT, SQL_BIGINT, or INTEGER64	REAL, SQL_REAL, or FLOAT32; DOUBLE PRECISION, SQL_DOUBLE, or FLOAT64; FLOAT, or SQL_FLOAT; EFLOAT	DECIMAL, DEC, NUMERIC, SQL_DECIMAL, or SQL_NUMERIC; DOLLAR	CHARACTER, ECHARACTER, CHARACTER VARYING, NATIONAL CHARACTER, NATIONAL CHARACTER VARYING, NLSCHARACTER, CHARACTER LARGE OBJECT, TEXT, NATIONAL CHARACTER LARGE OBJECT, NLSTEXT	BINARY LARGE OBJECT or BLOB; BULK	DATE, EDATE, TIME, ETIME, EPOCH_TIME, TIMESTAMP, MICROTIMESTAMP	BOOLEAN	SEQUENCE 32, SE
EXASolution	Static	TINYINT, SMALLINT, INTEGER, BIGINT,	REAL, FLOAT, DOUBLE	DECIMAL, DEC, NUMERIC, NUMBER	CHAR, NCHAR, VARCHAR, VARCHAR2, NVARCHAR, NVARCHAR2, CLOB, NCLOB	N/A	DATE, TIMESTAMP, INTERVAL	BOOLEAN, BOOL	GEOMETR
FileMaker[101]	Static	Not Supported	Not Supported	NUMBER	TEXT	CONTAINER	TIMESTAMP	Not Supported	
Firebird[102]	?	INT64, INTEGER, SMALLINT	DOUBLE, FLOAT	DECIMAL, NUMERIC, DECIMAL(18, 4), DECIMAL(10, 4)	BLOB, CHAR, CHAR(x) CHARACTER SET UNICODE_FSS, VARCHAR(x) CHARACTER SET UNICODE_FSS, VARCHAR	BLOB SUB_TYPE TEXT, BLOB	TIMESTAMP	CHAR(1), INTEGER	TIMESTAMP, CH
HSQLDB[103]	Static	TINYINT (8-bit), SMALLINT (16-bit), INTEGER (32-bit), BIGINT (64-bit)	DOUBLE (64-bit)	DECIMAL, NUMERIC	CHAR, VARCHAR, LONGVARCHAR, CLOB	BINARY, VARBINARY, LONGVARBINARY, BLOB	DATE, TIME, TIMESTAMP, INTERVAL	BOOLEAN	OTHER (object), VARYING, AR
Informix Dynamic Server[104]	Static	SMALLINT (16-bit), INT (32-bit), INT8 (64-bit proprietary), BIGINT (64-bit)	SMALLFLOAT (32-bit), FLOAT (64-bit)	DECIMAL (32 digits float/fixed), MONEY	CHAR, VARCHAR, NCHAR, NVARCHAR, LVARCHAR, CLOB, TEXT	TEXT, BYTE, BLOB, CLOB	DATE, DATETIME, INTERVAL	BOOLEAN	SET, LIST, MULTIS TIMESERIES, SPAT BSON, USER DEFIN
Ingres[105]	Static	TINYINT (8-bit), SMALLINT (16-bit), INTEGER (32-bit), BIGINT (64-bit)	FLOAT4 (32-bit), FLOAT (64-bit)	DECIMAL	C, CHAR, VARCHAR, LONG VARCHAR, NCHAR, NVARCHAR, LONG NVARCHAR, TEXT	BYTE, VARBYTE, LONG VARBYTE (BLOB)	DATE, ANSIDATE, INGRESDATE, TIME, TIMESTAMP, INTERVAL	N/A	MONEY, OBJEC TABLE_KEY, USER DATA TYPES (vi
Linter SQL RDBMS	Static + Dynamic (in stored procedures)	SMALLINT (16-bit), INTEGER (32-bit), BIGINT (64-bit)	REAL(32-bit), DOUBLE(64-bit)	DECIMAL, NUMERIC	CHAR, VARCHAR, NCHAR, NVARCHAR, BLOB	BYTE, VARBYTE, BLOB	DATE	BOOLEAN	GEOMETRY, EX
MariaDB[106]	Static	TINYINT (8-bit), SMALLINT (16-bit), MEDIUMINT (24-bit), INT (32-bit), BIGINT (64-bit)	FLOAT (32-bit), DOUBLE (aka REAL) (64-bit)	DECIMAL	CHAR, BINARY, VARCHAR, VARBINARY, TEXT, TINYTEXT, MEDIUMTEXT, LONGTEXT	TINYBLOB, BLOB, MEDIUMBLOB, LONGBLOB	DATETIME, DATE, TIMESTAMP, YEAR	BIT(1), BOOLEAN (aka BOOL) = synonym for TINYINT	ENUM, SET, GIS (Geometry, Point LineString, Surface GeometryCollection, MultiCurve, MultiL MultiSurface, Mult
Microsoft SQL Server[107]	Static	TINYINT, SMALLINT, INT, BIGINT	FLOAT, REAL	NUMERIC, DECIMAL, SMALLMONEY, MONEY	CHAR, VARCHAR, TEXT, NCHAR, NVARCHAR, NTEXT	BINARY, VARBINARY, IMAGE, FILESTREAM	DATE, DATETIMEOFFSET, DATETIME2, SMALLDATETIME, DATETIME, TIME	BIT	CURSOR, TIME HIERARCHY UNIQUEIDENT SQL_VARIANT, XM Geometry, Geog
Microsoft SQL Server Compact (Embedded Database)[108]	Static	TINYINT, SMALLINT, INT, BIGINT	FLOAT, REAL	NUMERIC, DECIMAL, MONEY	NCHAR, NVARCHAR, NTEXT	BINARY, VARBINARY, IMAGE	DATETIME	BIT	TIMESTAMP, ROW UNIQUEIDENT IDENTITY, ROWC
MonetDB	Static	TINYINT, SMALLINT, INT, BIGINT	FLOAT, REAL, DOUBLE	NUMERIC, DECIMAL	VARCHAR(n),CHAR(n),CLOB	BLOB	DATE, TIME, DATETIME, TIMESTAMP	BIT	TIME WITH TIM TIMESTAMP WI ZONE, DAY, MONT INTERVA
MySQL[98]	Static	TINYINT (8-bit), SMALLINT (16-bit), MEDIUMINT (24-bit), INT (32-bit), BIGINT (64-bit)	FLOAT (32-bit), DOUBLE (aka REAL) (64-bit)	DECIMAL	CHAR, BINARY, VARCHAR, VARBINARY, TEXT, TINYTEXT, MEDIUMTEXT, LONGTEXT	TINYBLOB, BLOB, MEDIUMBLOB, LONGBLOB	DATETIME, DATE, TIMESTAMP, YEAR	BIT(1), BOOLEAN (aka BOOL) = synonym for TINYINT	ENUM, SET, GIS (Geometry, Point LineString, Surface GeometryCollection, MultiCurve, MultiL MultiSurface, Mult

	Type system	Integer	Floating point	Decimal	String	Binary	Date/Time	Boolean	Other
OpenLink Virtuoso[109]	Static + Dynamic	INT, INTEGER, SMALLINT	REAL, DOUBLE PRECISION, FLOAT, FLOAT('INTNUM')'	DECIMAL, DECIMAL('INTNUM')', DECIMAL('INTNUM', 'INTNUM')', NUMERIC, NUMERIC('INTNUM')', NUMERIC('INTNUM', 'INTNUM')'	CHARACTER, CHAR('INTNUM')', VARCHAR, VARCHAR('INTNUM')', NVARCHAR, NVARCHAR('INTNUM')'	BLOB	TIMESTAMP, DATETIME, TIME, DATE	n/a	ANY, REFERENCE UDT (User Defin GEOMETRY (BOX BOX3D, BOXM, BOXZM, CIRCULA COMPOUNDC CURVEPOLYGON GEOMETRYCOLL GEOMETRYCOLL GEOMETRYCOLL GEOMETRYCOLLE LINESTRING, LINE LINESTRIN LINESTRINC MULTICUR MULTILINEST MULTILINESTF MULTILINESTI MULTILINESTR MULTIPOINT, MUL MULTIPOIN MULTIPOINT MULTIPOLY MULTIPOLYG MULTIPOLYG MULTIPOLYGONZ POINTM, POINTZ, POLYGON, POLY POLYGONZ, POLY POLYLINE, POLYLI RINGM, RINGZ, F
Oracle[110]	Static + Dynamic (through ANYDATA)	NUMBER	BINARY_FLOAT, BINARY_DOUBLE	NUMBER	CHAR, VARCHAR2, CLOB, NCLOB, NVARCHAR2, NCHAR, LONG (deprecated)	BLOB, RAW, LONG RAW (deprecated), BFILE	DATE, TIMESTAMP (with/without TIMEZONE), INTERVAL	N/A	SPATIAL, IMAGE VIDEO, DICOM,
Pervasive PSQL[111]	Static	BIGINT, INTEGER, SMALLINT, TINYINT, UBIGINT, UINTEGER, USMALLINT, UTINYINT	BFLOAT4, BFLOAT8, DOUBLE, FLOAT	DECIMAL, NUMERIC, NUMERICSA, NUMERICSLB, NUMERICSLS, NUMERICSTB, NUMERICSTS	CHAR, LONGVARCHAR, VARCHAR	BINARY, LONGVARBINARY, VARBINARY	DATE, DATETIME, TIME	BIT	CURRENCY, IDI SMALLIDENT TIMESTAM UNIQUEIDENT
Polyhedra[112]	Static	INTEGER8 (8-bit), INTEGER(16-bit), INTEGER (32-bit), INTEGER64 (64-bit)	FLOAT32 (32-bit), FLOAT (aka REAL; 64-bit)	N/A	VARCHAR, LARGE VARCHAR (aka CHARACTER LARGE OBJECT)	LARGE BINARY (aka BINARY LARGE OBJECT)	DATETIME	BOOLEAN	N/A
PostgreSQL[113]	Static	SMALLINT (16-bit), INTEGER (32-bit), BIGINT (64-bit)	REAL (32-bit), DOUBLE PRECISION (64-bit)	DECIMAL, NUMERIC	CHAR, VARCHAR, TEXT	BYTEA	DATE, TIME (with/without TIMEZONE), TIMESTAMP (with/without TIMEZONE), INTERVAL	BOOLEAN	ENUM, POINT, LIN BOX, PATH, POI CIRCLE, CIDR MACADDR, BIT, U JSON, JSONB, composites, range
Raima Database Manager[114]	Static	tinyint, smallint, integer, bigint	real, float, double	N/A	char, varchar, wchar, varwchar, long varchar, long varwchar	binary, varbinary, long varbinary	date, time, timestamp	bit	N/A
RDM Server[115]	Static	tinyint, smallint, integer, bigint	real, float, double	decimal, numeric	char, varchar, wchar, varwchar, long varchar, long varwchar	binary, varbinary, long varbinary	date, time, timestamp	bit	rowid
SAP HANA	Static	TINYINT, SMALLINT, INTEGER, BIGINT	SMALLDECIMAL, REAL, DOUBLE, FLOAT, FLOAT(n)	DECIMAL	VARCHAR, NVARCHAR, ALPHANUM, SHORTTEXT	VARBINARY, BINTEXT, BLOB	DATE, TIME, SECONDDATE, TIMESTAMP	BOOLEAN	CLOB, NCLOB, TE ST_GEOMETRY, S ST_MULTIPC ST_LINESTR ST_MULTILINES ST_POLYG ST_MULTIPOL ST_GEOMETRYCOI ST_CIRCULARS
solidDB	Static	TINYINT (8-bit), SMALLINT (16-bit), INTEGER (32-bit), BIGINT (64-bit)	REAL (32-bit), DOUBLE (64-bit), FLOAT (64-bit)	DECIMAL, NUMERIC (51 digits)	CHAR, VARCHAR, LONG VARCHAR, WCHAR, WVARCHAR, LONG WVARCHAR	BINARY, VARBINARY, LONG VARBINARY	DATE, TIME, TIMESTAMP		
SQLite[116]	Dynamic	INTEGER (64-bit)	REAL (aka FLOAT, DOUBLE) (64-bit)	N/A	TEXT (aka CHAR, CLOB)	BLOB	N/A	N/A	N/A
SQream DB[117]	Static	TINYINT (8-bit), SMALLINT (16-bit), INTEGER (32-bit), BIGINT (64-bit)	REAL (32-bit), DOUBLE (aka FLOAT) (64-bit)	N/A	CHAR, VARCHAR, NVARCHAR	N/A	DATE, DATETIME (aka TIMESTAMP)	BOOL	N/A
Teradata	Static	BYTEINT (8-bit), SMALLINT (16-bit), INTEGER (32-bit), BIGINT (64-bit)	FLOAT (64-bit)	DECIMAL, NUMERIC (38 digits)	CHAR, VARCHAR, CLOB	BYTE, VARBYTE, BLOB	DATE, TIME, TIMESTAMP (w/wo TIMEZONE)		PERIOD, INTE GEOMETRY, XML, (User Defined
UniData	Dynamic	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
UniVerse	Dynamic	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Type system	**Integer**	**Floating point**	**Decimal**	**String**	**Binary**	**Date/Time**	**Boolean**	**Other**

Other objects

Information about what other objects are supported natively.

	Data Domain	Cursor	Trigger	Function [1]	Procedure [1]	External routine [1]
4th Dimension	Yes	No	Yes	Yes	Yes	Yes
ADABAS	?	Yes	?	Yes?	Yes?	Yes
Adaptive Server Enterprise	Yes	Yes	Yes	Yes	Yes	Yes
Advantage Database Server	Yes	Yes	Yes	Yes	Yes	Yes
Altibase	Yes	Yes	Yes	Yes	Yes	Yes
Apache Derby	No	Yes	Yes	Yes [2]	Yes [2]	Yes [2]
ClustrixDB	No	Yes	No	Yes	Yes	Yes
CUBRID	Yes	Yes	Yes	Yes	Yes [2]	Yes
Drizzle	Yes	Yes	Yes [4]	Yes [4]	Yes [4]	Yes [4]
Empress Embedded Database	Yes via RANGE CHECK	Yes	Yes	Yes	Yes	Yes
EXASolution	Yes	No	No	Yes	Yes	Yes
DB2	Yes via CHECK CONSTRAINT	Yes	Yes	Yes	Yes	Yes
Firebird	Yes	Yes	Yes	Yes	Yes	Yes
HSQLDB	Yes	No	Yes	Yes	Yes	Yes
H2	Yes	No	Yes [2]	Yes [2]	Yes [2]	Yes
Informix Dynamic Server	Yes via CHECK	Yes	Yes	Yes	Yes	Yes [5]
Ingres	Yes	Yes	Yes	Yes	Yes	Yes
InterBase	Yes	Yes	Yes	Yes	Yes	Yes
Linter SQL RDBMS	No	Yes	Yes	Yes	Yes	No
LucidDB	No	Yes	No	Yes [2]	Yes [2]	Yes [2]
MariaDB	No [3]	Yes	Yes	Yes	Yes	Yes
MaxDB	Yes	Yes	Yes	Yes	Yes	?
Microsoft Access (JET)	Yes	No	No	No	Yes, But single DML/DDL Operation	Yes
Microsoft Visual Foxpro	No	Yes	Yes	Yes	Yes	Yes
Microsoft SQL Server	Yes	Yes	Yes	Yes	Yes	Yes
Microsoft SQL Server Compact (Embedded Database)	No	Yes	No	No	No	No
MonetDB	No	No	Yes	Yes	Yes	Yes
MySQL	No [3]	Yes	Yes	Yes	Yes	Yes
OpenBase SQL	Yes	Yes	Yes	Yes	Yes	Yes
Oracle	Yes	Yes	Yes	Yes	Yes	Yes
Oracle Rdb	Yes	Yes	Yes	Yes	Yes	Yes
OpenLink Virtuoso	Yes	Yes	Yes	Yes	Yes	Yes
Pervasive PSQL	Yes	Yes	Yes	Yes	Yes	No
Polyhedra DBMS	No	No	Yes	Yes	Yes	Yes
PostgreSQL	Yes	Yes	Yes	Yes	Yes	Yes
Raima Database Manager	No	Yes	No	No	Yes	Yes
RDM Server	No	Yes	Yes	No	Yes	Yes
SAP HANA	?	?	?	?	?	?
ScimoreDB	No	No	No	No	Yes	Yes
solidDB	Yes	Yes	Yes	Yes	Yes	Yes
SQL Anywhere	Yes	Yes	Yes	Yes	Yes	Yes
SQLite	No	No	Yes	No	No	Yes
Teradata	No	Yes	Yes	Yes	Yes	Yes
UniData	No	No	Yes	Yes	Yes	Yes
UniVerse	No	No	Yes	Yes	Yes	Yes
	Data Domain	**Cursor**	**Trigger**	**Function [1]**	**Procedure [1]**	**External routine [1]**

Note (1): Both **function** and **procedure** refer to internal routines written in SQL and/or procedural language like PL/SQL. **External routine** refers to the one written in the host languages, such as C, Java, Cobol, etc. "Stored procedure" is a commonly used term for these routine types. However, its definition varies between different database vendors.

Note (2): In Derby, H2, LucidDB, and CUBRID, users code **functions** and **procedures** in Java.

Note (3): ENUM datatype exists. CHECK clause is parsed, but not enforced in runtime.

Note (4): In Drizzle the user codes **functions** and **procedures** in C++.

Note (5): Informix supports external functions written in Java, C, & C++.

Partitioning

Information about what partitioning methods are supported natively.

	Range	Hash	Composite (Range+Hash)	List	Expression
4th Dimension	?	?	?	?	?
ADABAS	?	?	?	?	?
Adaptive Server Enterprise	Yes	Yes	No	Yes	?
Advantage Database Server	No	No	No	No	?
Altibase	Yes	Yes	No	Yes	?
Apache Derby	No	No	No	No	?
ClustrixDB	Yes	No	No	No	No
CUBRID	Yes	Yes	No	Yes	?
IBM DB2	Yes	Yes	Yes	Yes	?
Empress Embedded Database	No	No	No	No	?
EXASolution	No	Yes	No	No	No
Firebird	No	No	No	No	?
HSQLDB	No	No	No	No	?
H2	No	No	No	No	?
Informix Dynamic Server	Yes	Yes	Yes	Yes	Yes
Ingres	Yes	Yes	Yes	Yes	?
InterBase	No	No	No	No	?
Linter SQL RDBMS	Yes	No	No	Yes	No
MariaDB	Yes	Yes	Yes	Yes	?
MaxDB	No	No	No	No	?
Microsoft Access (JET)	No	No	No	No	?
Microsoft Visual Foxpro	No	No	No	No	?
Microsoft SQL Server	Yes	No	No	No	?
Microsoft SQL Server Compact (Embedded Database)	No	No	No	No	?
MonetDB	No	No	No	No	No
MySQL	Yes	Yes	Yes	Yes	?
OpenBase SQL	?	?	?	?	?
Oracle	Yes	Yes	Yes	Yes	?
Oracle Rdb	Yes	Yes	?	?	?
OpenLink Virtuoso	Yes	Yes	Yes	Yes	Yes
Pervasive PSQL	No	No	No	No	No
Polyhedra DBMS	No	No	No	No	No
PostgreSQL	Yes[1]	Yes[1]	Yes[1]	Yes[1]	?
Raima Database Manager	Yes[2]	Yes[2]	Yes[2]	No	?
RDM Server	No	No	No	No	?
SAP HANA	Yes	Yes	Yes	Yes	Yes
ScimoreDB	No	Yes	No	No	?
solidDB	Yes	No	No	No	?
SQL Anywhere	No	No	No	No	?
SQLite	No	No	No	No	?
Teradata	Yes	Yes	Yes	Yes	?
UniVerse	Yes	Yes	Yes	Yes	?
	Range	**Hash**	**Composite (Range+Hash)**	**List**	**Expression**

Note (1): PostgreSQL 8.1 provides partitioning support through check constraints. Range, List and Hash methods can be emulated with PL/pgSQL or other procedural languages.[118]

Note (2): Raima Database Manager 14.0 requires the application programs to select the correct partition (using range, hash or composite techniques) when adding data, but the *database union* functionality allows all partitions to be read as a single database.[119]

Access control

Information about access control functionalities (*work in progress*).

	Native network encryption[1]	Brute-force protection	Enterprise directory compatibility	Password complexity rules[2]	Patch access[3]	Run unprivileged[4]	Audit	Resource limit	Separation of duties (RBAC)[5]	Security Certification
4D	Yes (with SSL)	?	Yes	?	Yes	Yes	?	?	?	?
Adaptive Server Enterprise	Yes (optional; to pay)	Yes	Yes (optional ?)	Yes	Partial (need to register; depend on which product)[120]	Yes	Yes	Yes	Yes	Yes (EAL4+ [1])
Advantage Database Server	Yes	No	No	No	Yes	Yes	No	No	Yes	?
DB2	Yes	?	Yes (LDAP, Kerberos…)	Yes	?	Yes	Yes	Yes	Yes	Yes (EAL4+[6])
Empress Embedded Database	?	?	No	No	Yes	Yes	Yes	No	Yes	No
EXASolution	No	No	Yes (LDAP)	No	Yes	Yes	Yes	Yes	Yes	No
Firebird	No	Yes[121]	Yes (Windows trusted authenification)	No	Partial (no security page)[122]	Yes	Yes[123]	No	No[7]	?
HSQLDB	Yes	No	Yes	Yes	Yes	Yes	No	No	Yes	No
H2	Yes	Yes	?	No	?	Yes	?	Yes	Yes	No
Informix Dynamic Server	Yes	?	Yes[10]	?[10]	Yes	Yes	Yes	Yes	Yes	?
Linter SQL RDBMS	Yes (with SSL)	Yes	Yes	Yes (length only)	Yes	Yes	Yes	Yes	Yes	Yes
MariaDB	Yes (SSL)	No	Yes (with 5.2, but not on Windows servers)	Yes[124][125]	Yes[126]	Yes	?	?	?[8]	No
Microsoft SQL Server	Yes	?	Yes (Microsoft Active Directory)	Yes	Yes	Yes	Yes (From 2008)	Yes	Yes	Yes (EAL4+[11])
Microsoft SQL Server Compact (Embedded Database)	No (not relevant, only file permissions)	No (not relevant)	No (not relevant)	No (not relevant)	Yes	Yes (file access)	Yes	Yes	No	?
MySQL	Yes (SSL with 4.0)	No	Yes (with 5.5, but only in commercial edition)	No	Partial (no security page)[127]	Yes	?	?	?[8]	Yes
OpenBase SQL	Yes	?	Yes (Open Directory, LDAP)	No	?	?	?	?	?	?
OpenLink Virtuoso	Yes	Yes	Yes	Yes (optional)	Yes (optional)	Yes	Yes (optional)	Yes (optional)	Yes	No
Oracle	Yes	Yes	Yes	Yes	?	Yes	Yes	Yes	Yes	Yes (EAL4+[1])
Pervasive PSQL	Yes	?	No	No	Yes	Yes	Yes [12]	No	No	No
Polyhedra DBMS	Yes (with SSL. Optional)	No	No	No	No	Yes	Yes [13]	Yes	Yes [13]	No
PostgreSQL	Yes	Yes (for 9.1)	Yes (LDAP, Kerberos…[9])	Yes (as of 9.0 with passwordcheck module)	Yes[128]	Yes	No	Yes	Yes	Yes (EAL1[1])
Raima Database Manager	No	No	No	No	No	Yes	No	No	No	No
RDM Server	Yes	No	No	No	No	Yes	Yes	No	Yes	No
SAP HANA	?	?	?	?	?	?	?	?	?	?
solidDB	No	No	Yes	No	No	Yes	Yes	No	No	No
SQL Anywhere	Yes	?	Yes (Kerberos)	Yes	?	Yes	Yes	No	Yes	Yes (EAL3+[1] as Adaptive Server Anywhere)

	Native network encryption[1]	Brute-force protection	Enterprise directory compatibility	Password complexity rules[2]	Patch access[3]	Run unprivileged[4]	Audit	Resource limit	Separation of duties (RBAC)[5]	Security Certification
SQLite	No (not relevant, only file permissions)	No (not relevant)	No (not relevant)	No (not relevant)	Partial (no security page)[129]	Yes (file access)	Yes	Yes	No	No
Teradata	Yes	No	Yes (LDAP, Kerberos…)	Yes	?	Yes	Yes	Yes	Yes	Yes
	Native network encryption[1]	**Brute-force protection**	**Enterprise directory compatibility**	**Password complexity rules[2]**	**Patch access[3]**	**Run unprivileged[4]**	**Audit**	**Resource limit**	**Separation of duties (RBAC)[5]**	**Security Certification**

Note (1): Network traffic could be transmitted in a secure way (not clear-text, in general SSL encryption). Precise if option is default, included option or an extra modules to buy.

Note (2): Options are present to set a minimum size for password, respect complexity like presence of numbers or special characters.

Note (3): How do you get security updates? Is it free access, do you need a login or to pay? Is there easy access through a Web/FTP portal or RSS feed or only through offline access (mail CD-ROM, phone).

Note (4): Does database process run as root/administrator or unprivileged user? What is default configuration?

Note (5): Is there a separate user to manage special operation like backup (only dump/restore permissions), security officer (audit), administrator (add user/create database), etc.? Is it default or optional?

Note (6): Common Criteria certified product list.[130]

Note (7): FirebirdSQL seems to only have SYSDBA user and DB owner. There are no separate roles for backup operator and security administrator.

Note (8): User can define a dedicated backup user but nothing particular in default install.[131]

Note (9): Authentication methods.[132]

Note (10): Informix Dynamic Server supports PAM and other configurable authentication. By default uses OS authentication.

Note (11): Authentication methods.[133]

Note (12): With the use of Pervasive AuditMaster.

Note (13): User-based security is optional in Polyhedra, but when enabled can be enhanced to a role-based model with auditing.[134]

Databases vs schemas (terminology)

The SQL specification makes clear what an "SQL schema" is; however, different databases implement it incorrectly. To compound this confusion the functionality can, when incorrectly implemented, overlap with that of the parent-database. An SQL schema is simply a namespace within a database, things within this namespace are addressed using the member operator dot "`.`". This seems to be a universal amongst all of the implementations.

A true fully (database, schema, and table) qualified query is exemplified as such: `SELECT * FROM database.schema.table`

Now, the issue, both a schema and a database can be used to isolate one table, "foo" from another like named table "foo". The following is pseudo code:

- `SELECT * FROM db1.foo` vs. `SELECT * FROM db2.foo` (no explicit schema between db and table)
- `SELECT * FROM [db1.]default.foo` vs. `SELECT * FROM [db1.]alternate.foo` (no explicit db prefix)

The problem that arises is that former MySQL users will create multiple databases for one project. In this context, MySQL databases are analogous in function to Postgres-schemas, insomuch as Postgres lacks off-the-shelf cross-database functionality that MySQL has. Conversely, PostgreSQL has applied more of the specification implementing cross-table, cross-schema, and then left room for future cross-database functionality.

MySQL aliases *schema* with *database* behind the scenes, such that `CREATE SCHEMA` and `CREATE DATABASE` are analogs. It can therefore be said that MySQL has implemented cross-database functionality, skipped schema functionality entirely, and provided similar functionality into their implementation of a database. In summary, Postgres fully supports schemas but lacks some functionality MySQL has with databases, while MySQL does not even attempt to support true schemas.

Oracle has its own spin where creating a user is synonymous with creating a schema. Thus a database administrator can create a user called PROJECT and then create a table PROJECT.TABLE. Users can exist without schema objects, but an object is always associated with an owner (though that owner may not have privileges to connect to the database). With the Oracle 'shared-everything' RAC architecture, the same database can be opened by multiple servers concurrently. This is independent of replication, which can also be used, whereby the data is copied for use by different server. In the Oracle view, the 'database' is a set of files which contains the data while the 'instance' is a set of processes (and memory) through which a database is accessed.

Informix supports multiple databases in a server instance, like MySQL. It supports the CREATE SCHEMA syntax as a way to group DDL statements into a single unit creating all objects created as a part of the schema as a single owner. Informix supports a database mode called ANSI mode which supports creating objects with the same name but owned by different users.

The end result is confusion between the database factions. The Postgres and Oracle communities maintain that one database is all that is needed for one project, per the definition of database. MySQL and Informix proponents maintain that schemas have no legitimate purpose when the functionality can be achieved with databases. Postgres adheres to the SQL specification, in a more intuitive fashion (bottom-up), while MySQL's pragmatic counterargument allows their users to get the job done while creating conceptual confusion.

PostgreSQL and some other databases have recently added support for foreign schemas, the ability to import schemas from other servers, as defined in ISO/IEC 9075-9 (published as part of SQL:2008). This appears like any other schema in the database according to the SQL specification while accessing data stored either in a different database or a different server instance. The import can be made either as an entire foreign schema or merely certain tables belonging to that foreign schema.[135] While support for ISO/IEC 9075-9 bridges the gap between the two competing philosophies surround schemas, MySQL and Informix maintain an implicit association between databases while ISO/IEC 9075-9 requires that any such linkages be explicit in nature.

See also

- Relational database management system (includes market share data)
- List of relational database management systems
- Comparison of object-relational database management systems
- Comparison of database tools
- Object Database - some of which have relational (SQL/ODBC) interfaces.

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